

Script Function:

The script begins by generating a main directory named using the current date and time to ensure uniqueness. A log file named script.log is initialized to record all actions performed during execution. Ten subdirectories, named file101 through file110, are created within the main directory. Inside each subdirectory, ten text files named tuser501.txt through tuser510.txt are generated. Each file contains a unique programming language name. All directory creation, file creation, and verification actions are logged with timestamps, providing a complete execution record.

Purpose:

The purpose of this script is to automate the creation of directories and text files while maintaining a detailed execution log.

Linux Command Overview:

cp (Copy)

Copies files or directories from one location to another.

Example:

```
cp file.txt backup.txt
```

ps (Process Status)

Displays information about currently running processes.

Example:

```
ps aux
```

ls (List)

Lists files and directories in the current directory.

Example:

```
ls -l
```

mv (Move)

Moves or renames files or directories.

Example:

```
mv old.txt new.txt
```

rm (Remove)

Deletes files or directories.

Example:

```
rm file.txt
```

mkdir (Make Directory)

Creates a new directory.

Example:
mkdir myfolder

rmdir (Remove Directory)
Removes empty directories.

Example:
rmdir emptyfolder

pwd (Print Working Directory)
Displays the full path of the current directory.

Example:
pwd

stat (File Status)
Shows detailed information about a file including size, permissions, and timestamps.

Example:
stat script.log

tar (Archive Files)
Creates or extracts archive files.

Example:
tar -czf files.tar.gz folder/

apt (Package Manager)
Manages software packages on Debian-based systems such as Ubuntu.

Example:
sudo apt update

echo
Outputs text to the terminal or writes text to a file.

Example:
echo "Hello World"

cat (Concatenate)
Displays the contents of a file.

Example:
cat script.log

more
Views file contents one page at a time.

Example:
more script.log

less
Displays file contents with scrolling and search capabilities.

Example:
less script.log

date
Displays or formats the current system date and time.

Example:
date +"%Y-%m-%d"

time
Measures how long a command takes to execute.

Example:
time ./create_files_with_subdirs.sh

kill
Sends a signal to terminate or control a running process.

Example:
kill 1234

history
Displays a list of previously executed commands.

Example:
history

chmod (Change Mode)
Changes file permissions such as read, write, and execute.

Example:
chmod +x create_files_with_subdirs.sh

chown (Change Owner)
Changes the owner and group of a file.

Example:
sudo chown user:user script.log